RAL Space

Space Radiation Monitors

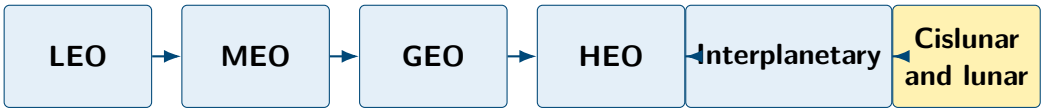
Heritage systems, emerging technologies and engineering gaps

Dr Ruth Bamford | RAL Space

State-of-the-art review update, 2026

The review connects mature markets to emerging missions

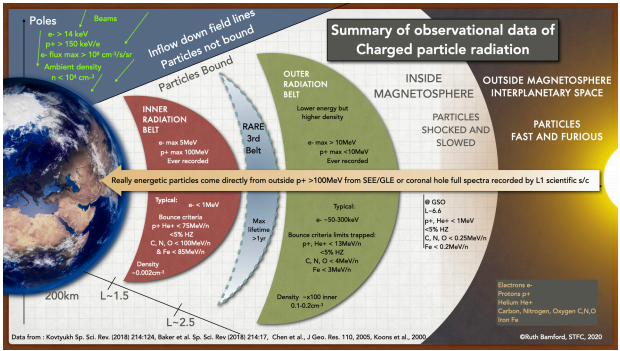
The survey is organised around the engineering question: **what measurement capability fits the mission?**



- The Kellett/SWIMMR material provides the heritage baseline.
- Post-2023 developments are compared separately, using the same engineering fields.
- Cislunar systems are included as an emerging market and a source of detector innovation.

Mission domain changes the useful energy range

- **LEO:** trapped particles, SAA passages and atmospheric secondaries.
- **MEO/GEO:** persistent energetic electrons, charging and solar-particle exposure.
- **Beyond Earth:** SEP, GCR, heavy ions and locally generated secondaries.



Engineering consequence:
No single energy range or detector architecture is optimal across every domain.

Detector families occupy different parts of the trade space

Family	Principal strength	Principal limitation
Silicon telescope	Heritage and ΔE–E identification	Thickness, geometry and contamination
Pixel / Timepix	Imaging, track classification and low SWaP	Stopping power and high-rate response
Scintillator	Active volume and high-energy stopping power	Species separation needs added information
Hybrid Si–scintillator	Tracking plus calorimetry over a broad range	Integration and cross-calibration complexity
Dosimetry / TEPC	Dose relevance and distributed monitoring	Limited spectral or directional information
Magnetic / ToF	Momentum, charge sign and composition	Mass, power and alignment

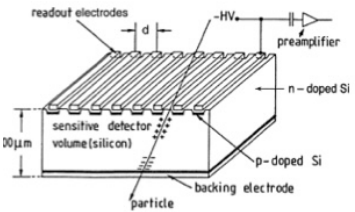
Heritage remains valuable ĂŤbut the instruments are not interchangeable

Operational heritage

Silicon telescopes and multi-sensor suites provide defensible calibration, species separation and wide energy coverage.

- SREM: multi-mission Earth-orbit and interplanetary heritage.
- NGRM/RMU: compact operational electron and proton/ion sensor heads.
- REPT: high-fidelity radiation-belt science.

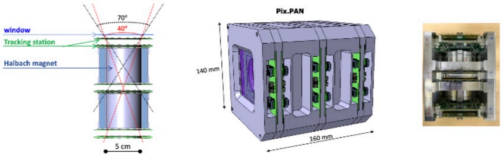
Solid-State Detectors (SSDs).jpeg



Miniaturised heritage

Timepix-based instruments add track imaging and distributed sensing, with different limits on stopping power and flux.

Post-2023 work expands the architecture, not just the catalogue



Pix.PAN combines magnetic analysis, Timepix4 tracking and time-of-flight.

- **MATCH:** DSSD and BGO stack for lunar particles.
- **PixPAN / NeuPix:** complementary charged- and neutral-particle concepts.
- **IRADCAL:** compact scintillator/SiPM system for lunar albedo studies.
- **HENON REPE:** directional electrons and protons from a 12U platform.

Common direction: more particle information within constrained accommodation.

The remaining gaps are system-level gaps

Capability	Present limitation	Opportunity
Broad-range low-SWaP	Selected bands or model-dependent unfolding	Hybrid and stacked architectures
Extreme SEP response	Pile-up, saturation and telemetry demand	Adaptive processing and dual gain
Compact neutron measurement	Often a separate payload	Converter pixels and segmented scintillators
Directional measurements	Acceptance competes with angular information	Stacked pixels and multiple sensor heads
Cross-mission comparison	Mission-specific response and products	Open response matrices and calibration metadata

The priority is not a universal sensor; it is a **cross-calibrated family of instruments**.

Select the measurement before selecting the detector

1. Define the operational or scientific data product.
2. Specify species, energy, cadence and dynamic range.
3. Test contamination, saturation and angular response.
4. Confirm mass, power, telemetry, interfaces and heritage.
5. Require a traceable calibration and response matrix.

Review conclusion

Pixel and hybrid systems broaden the available trade space, but they complement rather than automatically replace mature telescopes, calorimeters and dosimeters.

Next step

Complete the controlled comparison dataset and verify quantitative fields against primary sources.

Previous Presentation Tables

Orbit-Based View of Detector Developments

- Instruments are grouped by measurement environment:
  - heritage LEO mapping / dosimetry systems
  - MEO radiation-belt instruments
  - cislunar / lunar-environment concepts
- Highlighting used in the tables:
  - Red: not in the original baseline list
  - Blue: baseline instrument with updated / changed performance or status

LEO Instruments

Instrument	Type	Proton (MeV)	Mass	Status
HMRM	Pixel Si	0.6–500	52 g	flown
LUCID	Timepix array	>10	1.2 kg	flown
SATRAM	Timepix	ND	172–380 g	flown
MIRAM	Timepix3 + Si	10–400	>150 g	proposed / development
MERiT	SSD + APD	7–100	1.13 kg	flown
RADMON (Aalto-1)	Si + CsI	10–80	360 g	flown
RISEPix	2× Timepix stack	ND	>1 kg	flown
VZLUSAT-1	Timepix	ND	ND	flown
ALTEA	Si telescopes	ND	ND	flown
DosTel	Si telescope	ND	ND	flown
TEPC	Gas proportional counter	ND	ND	flown
HEPD	Si + scintillator	30–200	43 kg	flown

LEO remains dominated by miniaturised, mapping, and dosimetry-oriented systems, especially Timepix-class detectors.

MEO Instruments

Instrument	Type	Proton (MeV)	Mass	Status
EMU	Si telescope	15–100	3.5 kg	flown
Merlin	Si + SURF	>40	1.7 kg	flown
CEDEX	Si-based	ND	ND	flown
SREM (GIOVE-B)	Multi-detector	23–450	2.6 kg	flown
NGRM	Si (PSI)	2–200	1.4 kg	flown
ICARE-NG	Si coincidence	27–292	2.8 kg	flown
CEASE3	Si stacks	2–150	4 kg	flown
CREDANCE	Si detectors	ND	ND	flown
REPT	Si stack	17–200	13.4 kg	flown

MEO systems are primarily radiation-belt instruments with thicker silicon stacks and stronger spectral capability.

Cislunar / Lunar-Orbit Instruments

Instrument	Type	Energy Range	Platform	Status
CRaTER	Si pairs + tissue-equivalent plastic	ND	LRO lunar orbit	flown
RADOM	Mini dosimeter spectrometer	ND	Chandrayaan-1 lunar orbit	flown
MATCH	DSSD + BGO	MeV range	Chang'E-7	development
PixPAN	Magnetic spectrometer + Timepix4	0.1–10 GeV	ESA LunPAN	concept
NeuPix	Hybrid pixel detector	10–100 MeV	ESA LunPAN	concept
IRADCAL	Scintillator + SiPM	keV–100 MeV	2U CubeSat concept	concept

Cislunar developments are increasingly hybrid, with explicit focus on lunar albedo particles, neutrons and gamma-rays.

Comparison Across Orbit Regimes

Orbit Regime	Dominant Detector Style	Main Driver	Typ
LEO	Pixel / compact Si / dosimetry	mapping, dose, variability	lo
MEO	Silicon telescope stacks	trapped belts / spectra	high
Cislunar	Hybrid pixel + scintillator + spectrometer	SEP, GCR, albedo, neutrons	multi-sp

- LEO: compact deployment and distributed monitoring
- MEO: spectral discrimination in radiation belts
- Cislunar: broad environment, secondaries from regolith, neutron sensitivity

Physics Implications

- **LEO:** compact instruments are often sufficient for dose monitoring and spatial mapping
- **MEO:** thicker detector stacks are required to resolve trapped particle populations
- **Cislunar:** instrument requirements broaden to include:
  - SEP protons in the 10–100 MeV range
  - GCR response at higher energies
  - secondary neutrons and gamma-rays from regolith
- This is why recent cislunar concepts are increasingly hybrid rather than pure silicon telescope or pure pixel systems

Summary

- Grouping instruments by orbit clarifies both technology choice and physics driver
- Red entries identify instruments absent from the original baseline list
- Blue entries identify baseline instruments whose status / performance has changed
- LEO remains dominated by compact and pixel-based monitors
- MEO remains the domain of silicon telescope systems
- Cislunar developments show the clearest shift toward hybrid architectures